Chapter 1

INTRODUCTION

1.1 Introduction

The rapid digital transformation in the real estate industry has shifted the paradigm of property viewing from traditional physical walkthroughs to digital immersive experiences. Virtual tours have become an essential tool for property marketing, providing potential buyers with a spatial understanding of a property without the constraints of geographical distance or time. However, the creation of high-quality virtual tours remains largely inaccessible to independent brokers and small-scale agencies due to the high cost of specialized hardware, such as 360-degree cameras and LiDAR-equipped devices. This study addresses the democratization of immersive real estate technology by leveraging Computer Vision to reconstruct virtual tours from standard, multi-angle static images captured via common smartphone hardware.

1.2 Background of the Study

The evolution of virtual reality (VR) in real estate has transitioned from simple photo galleries to sophisticated, navigable environments. Currently, the gold standard for virtual tours involves the use of specialized 360-degree cameras or high-end mobile devices equipped with Light Detection and Ranging (LiDAR) sensors. While these technologies provide high-fidelity reconstructions, they impose significant financial and technical barriers. Independent brokers often find the procurement of such equipment or the hiring of professional photographers

cost-prohibitive, leading to a digital divide where only high-end listings benefit from immersive visualization.

Existing systems often rely on proprietary hardware or complex cloud-based stitching services that charge per-listing fees. Furthermore, mobile applications that attempt to simulate panoramic views often require the user to perform a continuous, steady sweep of the room, which is prone to motion blur and stitching artifacts. The rise of Computer Vision (CV) offers a viable alternative through automated image registration and stitching algorithms. By utilizing feature-based methods, it is possible to take multiple static images from varying angles and computationally align them to form a cohesive panoramic environment.

Despite the advancements in image processing, there remains a research gap in providing a localized, end-to-end platform that combines automated CV-based reconstruction with a dedicated real estate management system. Most solutions are either purely technical image-stitching tools or purely administrative listing platforms. There is a clear need for an integrated system that allows brokers to upload ordinary photos and automatically receive an immersive, web-accessible virtual tour. This study justifies the development of such a system to provide an accessible, low-cost, and mobile-responsive solution for the Philippine real estate market.

1.3 Objectives of the Study

1.3.1 General Objective

The primary objective of this study is to develop a computer vision-based virtual tour reconstruction system that enables the creation of immersive panoramic property tours using multi-angle static images captured from standard mobile devices.

1.3.2 Specific Objectives

1. To design and implement a web-based and mobile-responsive real estate platform using Next.js and Tailwind CSS.
2. To develop a backend processing engine using FastAPI and OpenCV for the automated reconstruction of virtual tours.
3. To implement an image preprocessing module capable of performing blur detection to ensure the quality of the input data.
4. To utilize Oriented FAST and Rotated BRIEF (ORB) for robust feature detection and description within the static images.
5. To apply Brute-Force Matcher (BFMatcher) and Random Sample Consensus (RANSAC) for accurate feature matching and homography estimation.
6. To perform image warping and blending to generate a seamless panoramic output from multiple overlapping frames.
7. To integrate the Marzipano library for rendering the stitched images into a navigable 360-degree virtual environment.
8. To develop a real-time messaging feature using Firebase to facilitate communication between property brokers and prospective buyers.
9. To provide a secure property listing management system for brokers to organize and showcase reconstructed tours.

10. To evaluate the system's performance and usability based on the ISO 25010 software quality standards, focusing on functionality, reliability, and user satisfaction.

1.4 Conceptual Framework

The development of the system is guided by the Input-Process-Output (IPO) model, supplemented by an Evaluation phase to ensure technical and functional rigor.

Input: The system requires primary data consisting of multi-angle static room images provided by the broker. These images must contain sufficient overlapping regions to allow for feature detection. Secondary inputs include property metadata (price, location, description), broker credentials, and user inquiries generated during buyer-broker interactions.

Process: The processing phase begins with image preprocessing, where images are converted and checked for quality via blur detection. The Computer Vision pipeline then executes ORB feature detection to identify unique keypoints in each image. These keypoints are matched across pairs using BFMatcher, followed by RANSAC to filter out outliers and calculate the homography matrix. Image warping aligns the perspectives, and panorama stitching blends the edges to form a single continuous image. This output is then rendered into an immersive interface using Marzipano, while the messaging and listing features are managed through Firebase's real-time database and authentication services.

Output: The primary output is a fully reconstructed, immersive virtual tour that users can navigate via a web browser. Secondary outputs include the published property listings visible on the public platform and a functional communication channel between buyers and brokers.

Evaluation: To validate the system, it will be subjected to testing based on the ISO 25010 Software Quality Model. This includes assessing Functional Suitability (does it stitch correctly?), Performance Efficiency (processing time), Usability (user interface ease of use), and Reliability (system uptime and error handling).

1.5 Definition of Terms

Computer Vision. Conceptually, it is a field of artificial intelligence that enables computers to derive meaningful information from digital images or videos (Szeliski, 2022). Operationally, it refers to the use of OpenCV libraries to automate the stitching of room images.

Panorama Stitching. The process of combining multiple photographic images with overlapping fields of view to produce a high-resolution panorama. In this study, it refers to the final blended output of the multi-angle room photos.

ORB (Oriented FAST and Rotated BRIEF). A fast robust feature detector used as an alternative to SIFT or SURF. It is used here to identify unique points in static images for alignment purposes.

Homography. A transformation (3x3 matrix) that maps the points in one image to the corresponding points in another image. It is the mathematical basis for aligning two images taken from different angles.

Virtual Tour. A simulation of an existing location, usually composed of a sequence of videos or still images. In this study, it refers to the interactive 360-degree view rendered through the Marzipano tool.

Feature Matching. The process of establishing correspondences between two sets of image features. It is operationally performed by the BFMatcher module in this system.

Immersive Visualization. A method of presenting information that surrounds the user, creating a sense of "being there." This refers to the interactive panoramic viewer on the web platform.

Static Images. Standard two-dimensional photographs captured by a smartphone. These serve as the raw data for the reconstruction process.

Real Estate Platform. A digital environment for the listing and sale of properties. Operationally, this is the Next.js application hosting the virtual tours and messaging features.

1.6 Significance of the Study

The completion of this study offers significant benefits to several stakeholders:

Real Estate Brokers and Agents. This study provides a cost-effective tool to enhance their digital presence. By removing the need for expensive cameras, individual agents can produce high-quality tours using their existing mobile devices, increasing their competitiveness in the market.

Property Buyers. Prospective buyers benefit from a more transparent and immersive viewing experience. It allows them to inspect properties remotely with greater spatial context than standard galleries provide, saving time and resources.

Real Estate Agencies. Smaller firms can adopt immersive technology at scale without significant capital expenditure, allowing for better standardization of listings across their organizations.

Researchers and Students. This study serves as a reference for applying Computer Vision techniques in a practical, consumer-facing application. It demonstrates the integration of modern web frameworks (Next.js, FastAPI) with complex image processing pipelines.

Future Developers. The architecture and methodologies documented in this research provide a foundation for further enhancements, such as 3D floor plan generation or AI-driven interior design staging.

1.7 Scope and Limitations

The study focuses on the development of a web-based real estate platform with an integrated Computer Vision engine for virtual tour reconstruction. The system includes features for user authentication, property listing management, multi-image upload, automated stitching, and a real-time messaging system. The platform is designed to be mobile-responsive, ensuring accessibility across desktops, tablets, and smartphones.

However, several limitations apply. The system is designed for interior room reconstruction and may not perform optimally for outdoor environments with infinite depth of field. The quality of the virtual tour is highly dependent on the user’s ability to capture overlapping images; the system requires a minimum overlap (typically 30-40%) between adjacent photos to successfully match features. While the system provides an immersive panoramic view, it does not perform a "perfect" 3D reconstruction or provide exact volumetric

measurements (e.g., floor area) as a LiDAR-based system would. Finally, the processing speed of the stitching is subject to the resolution of the uploaded images and the server-side hardware capabilities of the FastAPI backend.

Chapter 2

REVIEW OF RELATED LITERATURE

This chapter reviews relevant literature and studies related to Computer Vision applications in real estate, panorama reconstruction, image stitching, immersive visualization, feature detection algorithms, homography estimation, and low-cost virtual tour technologies. The review also identifies research gaps that justify the development of the proposed Computer Vision-Based Virtual Tour Reconstruction System using ordinary multi-angle smartphone images.

Computer Vision in Real Estate Applications

Computer Vision has become increasingly influential in Property Technology (PropTech), particularly in automating image analysis, property visualization, and spatial reconstruction (Matterport, 2024; Liu et al., 2022). According to Szeliski (2022), computer vision enables machines to interpret and process visual information through techniques such as feature extraction, image alignment, object recognition, and geometric transformations. These capabilities have expanded the possibilities for automated visualization systems within the real estate sector, as demonstrated by industry platforms that generate immersive 3D property tours (Matterport, 2024).

In the Philippine context, the computer vision market is projected to grow at a compound annual growth rate of 7.18% from 2026 to 2034, driven by digital transformation initiatives and investments in smart surveillance infrastructure (IMARC Group, 2026). However, the same report indicates that major deployment areas include manufacturing quality control, security

surveillance, and traffic monitoring, with no mention of immersive virtual tour creation for real estate applications. While virtual tour technologies have been deployed for tourism promotion in the Philippines (Department of Tourism & TPB, 2022), their application to real estate marketing—particularly for small-scale brokers—remains unexplored. This gap suggests an opportunity for low-cost virtual tour systems that do not require specialized hardware.

Virtual Tour Systems and Immersive Visualization

Immersive visualization significantly improves spatial understanding compared to fragmented 2D images. Research into Extended Reality (XR) demonstrates that AI-enhanced immersive environments bridge the gap between physical objects and user comprehension. In the context of the Philippines and other regions, university-led pilot studies have shown that web-based virtual tours using platforms like Story Spheres effectively stimulate user engagement and provide positive experiential value (Figueroa et al., 2020). Furthermore, virtual reality experiences are found to trigger positive emotional responses, such as pride and attachment, which are critical in high-stakes decision-making environments like real estate.

Chapter 3

METHODOLOGY

This chapter presents the comprehensive methodological framework and systematic procedures utilized in the development and evaluation of "LiBot". It details the iterative software development life cycle, the technical architectural design, and the computer vision pipeline employed to reconstruct immersive virtual tours. Furthermore, this section discusses the requirement analysis, the development environment, and the rigorous testing protocols based on international software quality standards. To ensure the system's reliability and practical utility, the researchers established data gathering procedures, identified appropriate respondents, and selected statistical tools to analyze the evaluative feedback. The methodology described herein serves as the blueprint for transforming ordinary multi-angle static images into a functional, web-responsive real estate platform.

Project Description

LiBot is a specialized web and mobile-responsive real estate application designed to democratize access to immersive property visualization. The primary purpose of the system is to solve the financial and technical barriers associated with traditional virtual tour creation, which typically requires expensive 360-degree cameras, LiDAR-equipped devices, or specialized panoramic hardware. By leveraging Computer Vision (CV) techniques, LiBot enables real estate brokers and agents to generate high-quality panoramic tours using ordinary smartphone cameras.

The system operates by allowing users to capture a series of overlapping static images of a room from various viewing angles. These images are then processed through a reconstruction engine that performs feature detection, matching, and seamless stitching to produce an equirectangular

panorama. This output is rendered through an immersive viewer, providing potential buyers with a 360-degree perspective of the property. Beyond visualization, the system integrates a robust real estate marketplace where brokers can manage listings and buyers can communicate directly through a built-in messaging module. LiBot emphasizes accessibility, affordability, and immersive visualization, providing a low-cost alternative for digital property marketing without compromising visual quality or user experience.

Software Development Methodology

The researchers utilized the Agile Software Development Methodology to facilitate the construction of LiBot. Agile was selected due to its iterative nature, allowing the researchers to refine the complex computer vision algorithms through continuous feedback and testing. Unlike linear models, Agile provides the flexibility required to adjust image stitching parameters and user interface components based on modular implementation.



Figure . Agile Software Development Methodology that serves as basis for developing LiBot Web and Mobile Application

Planning: In this initial phase, the researchers defined the project scope, identifying the specific computer vision libraries (OpenCV) and web frameworks (Next.js, FastAPI) needed. A roadmap was established to prioritize the development of the panorama reconstruction engine before the integration of marketplace features.

Requirement Gathering: The researchers identified the functional needs of brokers (image uploading, tour generation) and buyers (tour viewing, messaging). Non-functional requirements, such as processing speed for image stitching and mobile responsiveness, were also documented.

Design: This phase involved creating wireframes for the user interface and mapping the system architecture. Special attention was given to the immersive viewer's UI/UX to ensure intuitive navigation.

Development: The system was built in modular increments. The backend stitching engine was developed using FastAPI and OpenCV, while the frontend marketplace was constructed using Next.js. Integration with Firebase ensured real-time data handling for user authentication and messaging.

Testing: Each module underwent rigorous testing. The stitching engine was tested with various image sets to ensure homography accuracy, while the web platform was tested for cross-browser compatibility and mobile responsiveness.

Deployment: The system was deployed to a cloud environment, ensuring that the web application is accessible to users via standard browsers.

Maintenance: Post-deployment, the researchers monitored the system for bugs, particularly in the rendering of panoramas on different screen sizes, and applied patches as necessary to improve performance.

Requirement Analysis

The system requirements are categorized using the Input, Process, and Output (IPO) model to provide a clear overview of the data transformation within LiBot.

Input: The system accepts multi-angle static room images (JPEG/PNG) captured by brokers. Additionally, it requires broker profile data, property listing details (price,location, description), and buyer inquiries in the form of text messages.

Process: The processing core involves several technical stages. First, Image Preprocessing and Blur Detection ensure image quality. The system then performs ORB (Oriented FAST and Rotated BRIEF) Feature Detection to identify keypoints in the images. BFMatcher (Brute-Force Matcher) is used for feature matching across overlapping frames, followed by RANSAC (Random Sample Consensus) Homography Estimation to filter outliers and align images. The images undergo Image Warping and Panorama Stitching to create a seamless wide-angle view. Finally, the Marzipano library

handles Immersive Rendering, while Firebase manages the Messaging Process and database updates.

Output: The primary output is the Immersive Panoramic Virtual Tour, which reconstructs the room environment for 360-degree viewing. Secondary outputs include the generated property listings, stored messaging conversations, and a responsive visualization of the room.

The system is designed to meet high functional requirements, such as accurate tour generation, and non-functional requirements, including security (authorized access via Firebase), usability (intuitive UI), scalability (handling multiple concurrent uploads), and high performance (low latency in image processing).

Design Phase

The design of LiBot prioritizes a modern, clean interface that facilitates ease of navigation for both technical and non-technical users. Utilizing Tailwind CSS, the researchers ensured a UI Consistency across all pages. The Responsive Mobile Layout allows brokers to upload images directly from their phones on-site.

Login and Registration Page: Features secure authentication fields for brokers and buyers.

Broker Dashboard: A central hub for agents to track their active listings, view inquiry notifications, and initiate new virtual tour projects.

Image Upload Interface: A drag-and-drop or gallery-select interface that guides the broker through the multi-angle capture process.

Immersive Virtual Tour Viewer: A full-screen viewer using Marzipano that supports touch gestures and mouse navigation for 360-degree exploration.

Property Listing Page: Displays the reconstructed tour alongside property metadata and the broker's contact details.

Buyer Messaging Page: A real-time chat interface for direct communication.

Admin Dashboard: Provides tools for the system administrator to moderate listings and manage user accounts.

Environment

The development and deployment environment consists of specific hardware and software components optimized for computer vision processing and web delivery.

- **Hardware Requirements:**

  Processor: Intel Core i5/i7 (8th Gen or higher) or AMD Ryzen 5/7 for efficient image processing.

  RAM: Minimum 8GB (16GB recommended) to handle concurrent image stitching tasks.

  Storage: 256GB SSD for fast read/write operations.

  Smartphone Camera: Standard 12MP camera or higher for high-quality static image capture.

- **Software Requirements:**

Development Tools: Visual Studio Code, GitHub for version control.

Programming Languages: Python (Backend/CV), JavaScript/TypeScript (Frontend).

Libraries & Frameworks: OpenCV (CV Engine), FastAPI (API), Next.js (Web Framework), Marzipano (360 Rendering), Tailwind CSS (Styling).

Database & Auth: Firebase (Real-time DB, Storage, and Authentication).

- **Deployment Environment:**

  Browser Compatibility: Optimized for Google Chrome, Mozilla Firefox, and Safari.

  Connectivity: Requires a stable internet connection for image cloud processing.

Development Phase

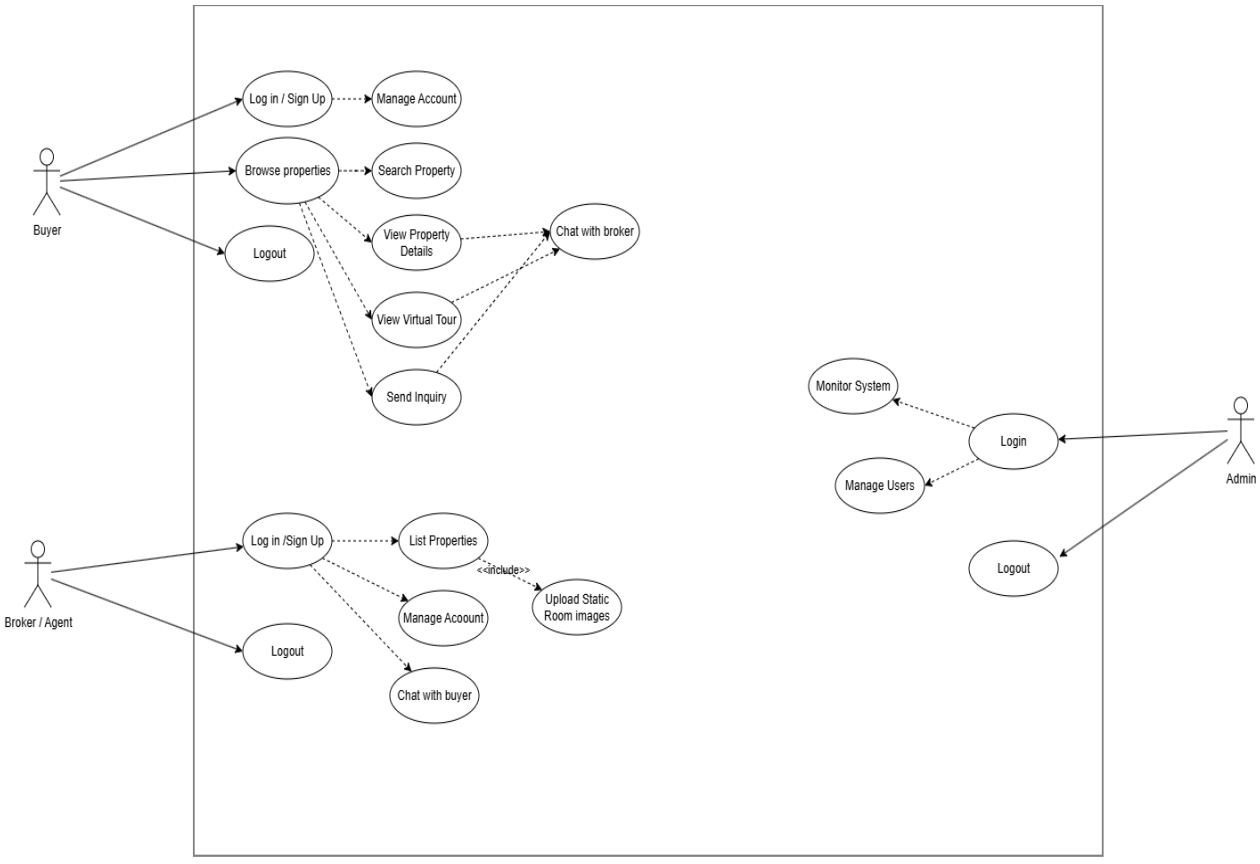
Following the validation of the prototype, the development phase of LiBot commenced. This stage entails the translation of the design specifications into a fully functional system using FastAPI for the backend engine and Next.js for the frontend. The programmer meticulously implemented the Computer Vision pipeline using the OpenCV library, focusing on the precision of the ORB (Oriented FAST and Rotated BRIEF) feature detection and BFMatcher logic.

The construction phase involved integrating Firebase for real-time database management and secure image storage. Adherence to industry-standard coding practices was paramount to ensure

the scalability and performance of the stitching engine. Throughout this phase, rigorous testing procedures were conducted to identify and rectify "ghosting" artifacts or seam misalignments in the panoramas. Continuous collaboration with technical advisers ensured the system's backend logic aligned with the project's goal of high-fidelity reconstruction.

Deployment Phase

Once the LiBot platform was fully developed, it was prepared for public access. This involves configuring the cloud hosting environment and ensuring the platform is optimized for the heavy computational load required for image processing. The researchers finalized the backend configurations to handle API requests from the mobile-responsive frontend. The system was then deployed to a web server, making it accessible via standard browsers on both desktop and mobile devices. Post-deployment, the site is monitored to ensure the Marzipano rendering engine maintains optimal frame rates during virtual tour navigation.



Project Design

Figure . Use Case Diagram of the LiBot System

1. Broker Actions:

1.1. Sign Up/Log In -

Access the system to manage property listings.

1.2. Upload Room Images -

Submit multi-angle static photos for CV processing.

1.3. Generate Virtual Tour -

Trigger the stitching engine to reconstruct the room.

1.4. Manage Accounts

- Edit or remove property details and tour views.

1.5. Messaging -

Respond to inquiries from potential buyers.

2. Buyer Actions:

2.1. Browse Listings -

Search for properties and view reconstructed tours.

2.2. Immersive Viewing -

Interact with the 360-degree virtual environment.

2.3. Send Inquiry - -

Contact brokers directly regarding a property listing.

3. Admin Actions:

3.1. User Management -

Verify broker credentials and monitor platform activity.

3.2. Content Moderation -

Review listings and reports to maintain platform integrity.

The system is designed to facilitate efficient interactions among three primary actors: property buyers, real estate brokers or agents, and system administrators. Each actor has specific roles and responsibilities that contribute to the overall functionality, reliability, and immersive experience of the LiBot platform.

- **Buyer:** Buyers can create an account, log in, and browse various property listings. The system allows them to experience properties remotely through immersive 360-degree virtual tours reconstructed from static images. Buyers can refine their searches using filters such as location, property type, and price range. Additionally, they have the ability to send direct inquiries and communicate with brokers via the integrated messaging feature to discuss property details or schedule site visits.
- **Broker / Agent:** Real estate brokers or agents are the primary content creators of the system. They can register and log in to post and manage property listings. Their unique role involves uploading ordinary, multi-angle static images of a room, which the system's Computer Vision engine then processes to generate an immersive panoramic tour. To

ensure platform trust, brokers are responsible for providing accurate property data and responding to buyer inquiries. They also manage their active listings and monitor the status of their virtual tour reconstructions.

- **Administrator:** The system administrator oversees all platform operations to maintain integrity, security, and technical performance. This includes reviewing and processing broker verification requests to prevent fraudulent listings. The administrator has the authority to approve, reject, or moderate property postings and evaluate reports from users to determine whether a listing or an account should be suspended for policy violations. Furthermore, the admin monitors the system's Computer Vision processing queues to ensure the efficient generation of virtual tours.

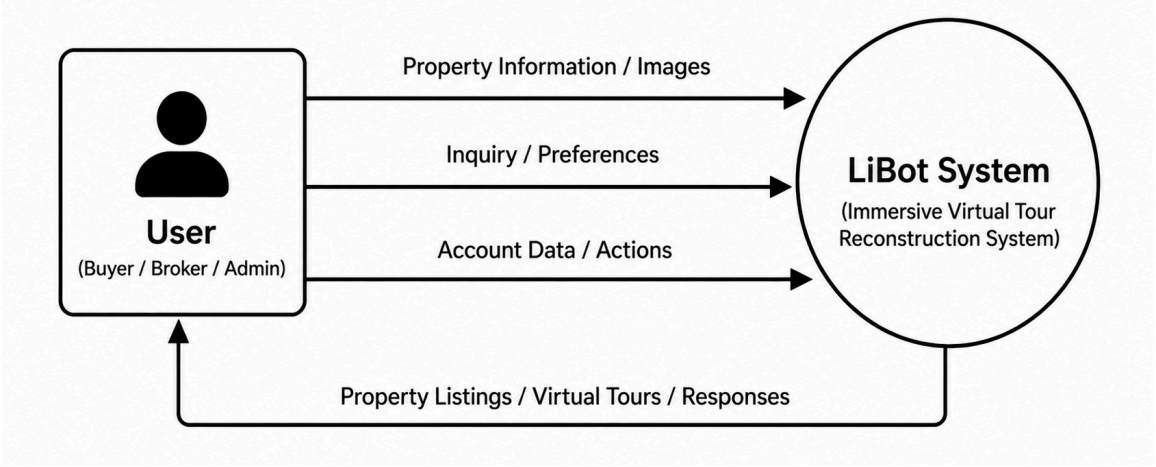


Figure 4. Context Diagram of the iLocate System

The context diagram illustrates the high-level interaction between the entities and the system. The Broker serves as a primary entity providing image data and listing details. These are processed by the LiBot System, which returns a generated virtual tour link. The Buyer inputs search preferences and inquiry messages, receiving immersive 360-degree visuals in return.

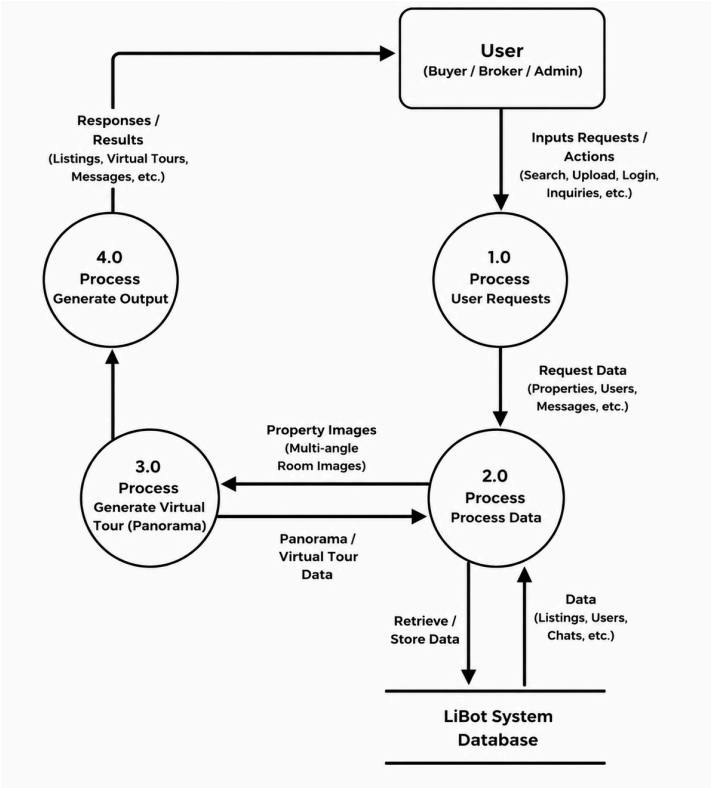


Figure 5. Level 0 Data Flow Diagram (DFD) of the ILOcate System

The Level 0 Data Flow Diagram illustrates the logical flow of data within the LiBot system, providing a detailed overview of how information is transformed from raw inputs into an immersive virtual environment. The diagram partitions the system into several core processes that manage user interactions, computer vision computations, and real-time communication.

The process begins with Process 1.0: Image and Property Management, where the Broker serves as the external entity providing raw multi-angle room images and property metadata (such as location, price, and amenities). This data is directed to Process 2.0: Computer Vision Reconstruction Engine, which serves as the system's primary computational component. In this

step, the system interacts with the OpenCV Library to perform high-level feature detection and extraction using the ORB algorithm. The module then executes BFMatcher to identify correspondences and applies RANSAC Homography Estimation to establish geometric relationships between the images. These data flows ensure that the raw pixels are accurately aligned and transformed into a unified equirectangular coordinate system.

Once the reconstruction logic is executed, the resulting stitched panorama is transmitted to Process 3.0: Media Storage and Rendering, where the file is archived within Firebase Storage. This process ensures that the high-resolution asset is mapped to the specific property listing and is ready to be fetched by the Immersive Viewer Module. Simultaneously, the Buyer interacts with the system by providing search criteria and preference data, which the system uses to retrieve the rendered panoramic tours via the Marzipano engine.

Furthermore, Process 4.0: Integrated Messaging System manages the bidirectional flow of inquiry data between the Buyer and the Broker. When a buyer sends an inquiry, the data is routed through this module, updated in the Real-time Database, and pushed as a notification to the broker's dashboard. This integration of complex computer vision processing with standard marketplace data flows allows LiBot to provide a seamless transition from static image capture to an interactive, preference-aware real estate experience. The DFD highlights the system's ability to maintain data integrity across cloud storage, local processing libraries, and front-end visualization interfaces.

```
START
    // Step 1: Input and Preprocessing
    INPUT Array of multi-angle images [I1, I2, ... In]
    FOR each image I in Array:
        Apply Grayscale conversion and Noise Reduction
        Detect Keypoints using ORB(I)
        Compute Descriptors using ORB(I)
    END FOR

    // Step 2: Feature Matching
    INITIALIZE BFMatcher
    MATCH descriptors between I(n) and I(n+1)
    FILTER matches using Lowe's Ratio Test to remove outliers

    // Step 3: Homography and Warping
    IF matches > threshold:
        ESTIMATE Homography Matrix (H) using RANSAC
        WARP Image I(n+1) onto the plane of I(n) using H
        STITCH warped image into a unified Canvas
    ELSE:
        RETURN Error "Insufficient overlap for reconstruction"
     END IF

    // Step 4: Blending and Rendering
    APPLY multi-band blending to smooth seams
    OUTPUT Stitched_Panorama.jpg
    RENDER via Marzipano for 360 viewer
END
```

Script 1: Pseudocode of Panorama Stitching Process

In the pseudocode shown in Illustration 1, the Panorama Stitching and Reconstruction process begins by taking a sequence of multi-angle static room images as input. The image data is first subjected to preprocessing and feature detection using the ORB (Oriented FAST and Rotated BRIEF) algorithm to identify unique keypoints and descriptors across the overlapping frames. Next, these descriptors are processed using the BFMatcher (Brute-Force Matcher) to establish accurate feature correspondences between adjacent images. These matches are then filtered using the RANSAC (Random Sample Consensus) algorithm to estimate the homography matrix, which ensures precise geometric alignment by eliminating outlier points.

Following alignment, the images undergo warping and stitching to be merged into a unified equirectangular canvas. Finally, the reconstructed panoramic image is output for rendering via the Marzipano engine, providing an immersive 360-degree virtual tour of the room. This process ensures high-fidelity environment reconstruction while maintaining computational efficiency and clarity in its implementation for web and mobile-responsive use.

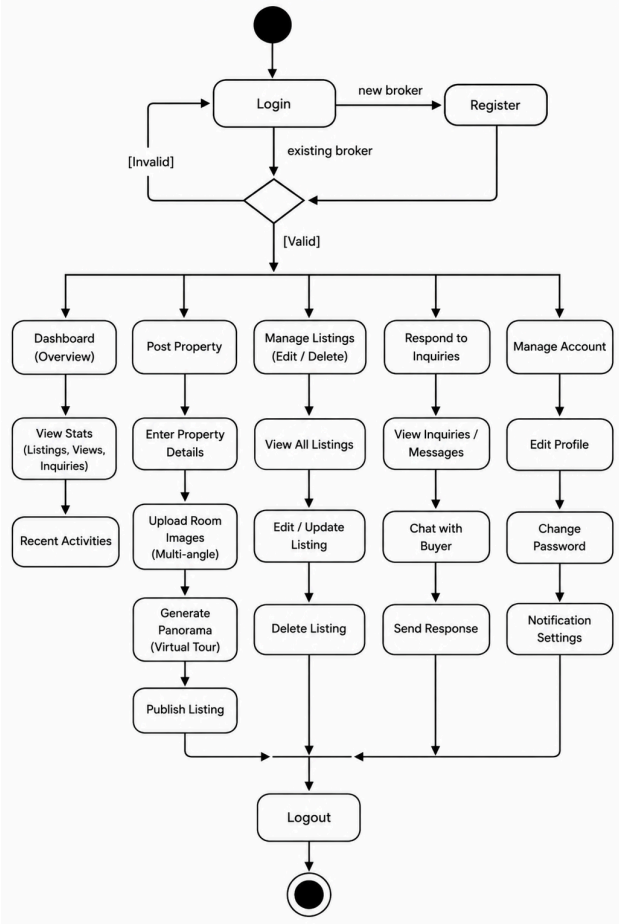


Figure 6. Activity Diagram of the Broker

1. **Login and Registration Process**

1.1. **Login**

- Existing brokers can log into the LiBot system using their registered username and password.
- If the account is not yet verified by the administrator, the broker is redirected to a pending status page.

- If the login credentials are invalid, the system prompts the user to retry. Upon a successful and verified login, the broker is granted access to the main dashboard.

1.2. **Registration (New Broker)**

- New users must register by providing their professional details, including their real estate license information or a valid ID for verification.
- Once the registration form is submitted, the account enters a "Pending" state until an administrator reviews and approves the credentials.
- After approval, the broker can proceed to log in and access the platform's features.

2. **Broker Main Activities**

Once authenticated, brokers can perform the following core operations:

2.1. **Post Property and Upload Images**

- Brokers input specific property data, such as location, price, and descriptions.
- The system prompts the broker to upload a series of multi-angle static images captured from a central pivot point in a room.
- These images are sent to the Computer Vision Reconstruction Engine to begin the panorama stitching process.

2.2. **Generate and Preview Virtual Tour**

- The broker triggers the "Generate Tour" action, which initiates the ORB feature detection and homography estimation.
- Once the system outputs the equirectangular panorama, the broker can preview the immersive 360-degree tour using the Marzipano viewer.
- If the reconstruction is satisfactory, the broker publishes the listing to make it visible to potential buyers.

2.3. **Manage Property Listings**

- Brokers can view a comprehensive list of their active and archived properties.
- The system allows for the modification of property details, the addition of new room tours, or the removal of sold properties from the public marketplace.

2.4. **Respond to Buyer Inquiries**

- Brokers receive real-time notifications when a buyer sends an inquiry regarding a listing.

- Through the integrated messaging interface, the broker can answer questions, provide additional details, and coordinate site visits or transactions.

2.5. **Manage Account**

- Allows the broker to update their professional profile, change login credentials, and manage contact information to ensure they remain reachable to clients.

3. **Logout Process**

- Brokers can securely log out of the platform, effectively terminating the session.
- This process clears session data to ensure the security of broker-client communications and protected property data.

The activity diagram for the broker illustrates a comprehensive workflow that transitions from account verification to the complex task of virtual tour reconstruction. The design emphasizes a logical progression from data input and computer vision processing to active marketplace management and client communication.

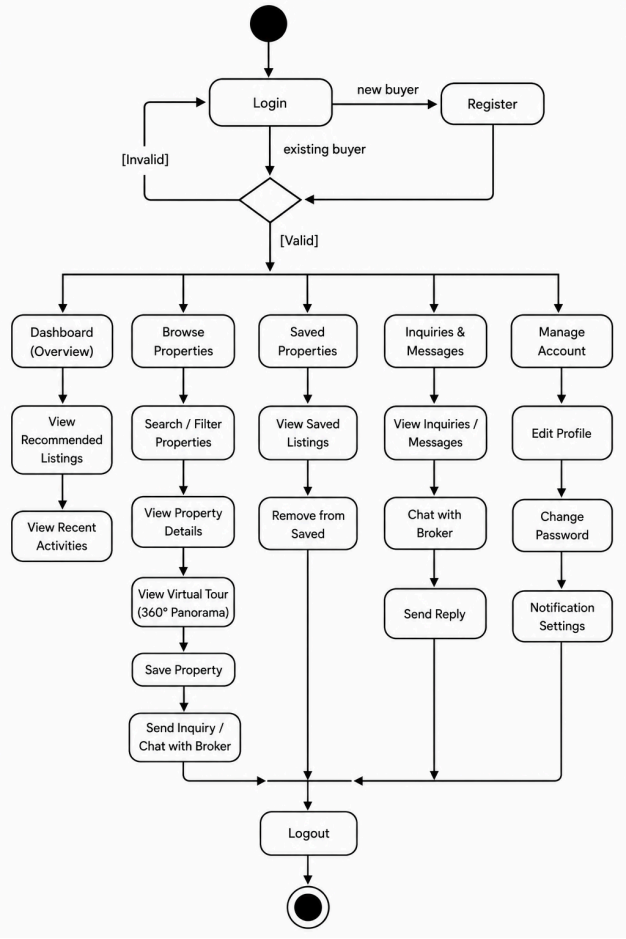


Figure 6. Activity Diagram of the Buyer

1. **Login and Registration Process**

1.1. **Login**

- Existing buyers can log into the system using their registered email and password.
- If the provided credentials do not match the records in the Firebase authentication database, the system generates an error message and prompts the user to retry.
- Upon valid authentication, the buyer is redirected to the home gallery, gaining access to the platform's full immersive functionalities.

1.2. **Registration (New Buyer)**

- New users must register by providing required account information such as name, contact details, and password.
- After the system validates the uniqueness of the account and stores the data, the new buyer can proceed to log in and begin exploring property listings.

2. **Buyer Main Activities**

Once logged in, the buyer can engage in the following primary interactions within the LiBot environment:

2.1. **Browse and Filter Property Listings**

- The buyer can view a comprehensive list of available properties posted by verified brokers.

- The system provides advanced filtering options, allowing buyers to navigate through listings based on specific preferences such as geographic location, property type, and price range.

2.2. **Experience Immersive Virtual Tours**

- Upon selecting a property, the buyer triggers the "View Tour" action.
- The system fetches the stitched equirectangular panorama from cloud storage and renders it through the Marzipano engine.
- The buyer can interact with the 360-degree environment using touch gestures or mouse controls, providing a realistic sense of the room's spatial layout and dimensions without being physically present.

2.3. **Send Inquiries and Messaging**

- While viewing a tour or property listing, the buyer can initiate contact with the broker by sending an inquiry.
- This opens a real-time messaging interface where the buyer can ask specific questions or request a site visit. All conversations are stored and can be accessed later via the buyer's message center.

2.4. **Manage Account and Favorites**

- Buyers can update their personal profile, manage notification settings, and review their history of viewed tours or saved property listings.
- This allows for a personalized experience, ensuring that the buyer can easily revisit properties of interest.

3. **Logout Process**

- The buyer can securely log out of their account, which terminates the active session and clears sensitive data from the browser's local state.
- This ensures the privacy of the buyer’s search history and personal communications, protecting their data from unauthorized access on shared devices.

The activity diagram for the buyer outlines a streamlined and user-centric flow that moves from standard account management to high-engagement visualization. The design ensures that the transition between browsing static listings and interacting with reconstructed 360-degree environments is seamless, intuitive, and technically responsive.

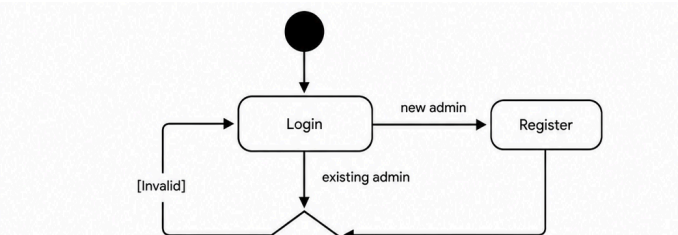


Figure 6. Activity Diagram of the Admin

1. **Login and Registration Process**

1.1. **Login**

- The administrator accesses the system through a dedicated administrative portal. Existing admins enter their pre-authorized credentials (username and password) which are validated against the Firebase administrative records.
- If the credentials are **invalid**, the system triggers a recursive loop back to the login screen with a security prompt.

- Upon **valid** authentication, the system initializes the admin session, granting access to the comprehensive suite of management modules.

1.2. **Registration**

- While administrative accounts are typically restricted, the system provides a registration flow for new administrative personnel.
- Once a new admin registers, the account must be cleared through a secondary security verification before they can participate in platform moderation.

2. **Admin Dashboard Activities**

Following a successful login, the administrator is presented with five core functional pillars designed for high-level platform oversight:

2.1. **Dashboard Overview and Statistics**

- The admin can monitor real-time system health, including statistics on active virtual tours and successful reconstruction rates.
- The dashboard displays recent activities, allowing the admin to track which brokers are most active and review comprehensive reports concerning users, listings, and system inquiries.

2.2. **User Management (Brokers and Buyers)**

- The administrator has the authority to view all registered users. This module allows for the activation or deactivation of broker accounts, which is critical for ensuring that only legitimate real estate professionals are generating tours.
- Admins can perform maintenance tasks such as resetting passwords and updating user roles to maintain organizational hierarchy within the platform.

2.3. **Property and Listing Management**

- This is a critical module for LiBot's quality control. The admin reviews all property listings and their corresponding reconstructed virtual tours.
- The admin holds the power to Approve or Reject listings based on the quality of the image stitching and the accuracy of the metadata. They can edit or remove listings that fail to meet the platform's immersive standards or community guidelines.

2.4. **Inquiries and Message Management**

- To ensure high service standards, the admin can view all inquiries and messages between buyers and brokers.

- They can assist in resolving disputes, closing resolved inquiries, or manually assigning inquiries to specific brokers if needed to ensure no potential buyer is left unassisted.

2.5. **System Settings and Maintenance**

- The administrator manages the platform's metadata, such as property categories, standard amenities, and location databases.
- This module also handles administrative notifications and critical technical tasks, including database backups and restoration to ensure the longevity and reliability of the virtual tour assets stored in Firebase.

3. **Logout Process**

- The administrator can securely log out of the system, which immediately invalidates the administrative token and terminates the session.
- This ensures that the high-level access granted to the admin dashboard is protected from unauthorized use, maintaining the overall security of the LiBot infrastructure.

The administrative activity flow transitions from a secure entry point to a multi-faceted dashboard that balances technical monitoring with community moderation. By overseeing both the users and the technical output of the Computer Vision engine (the virtual tours), the administrator ensures that LiBot remains a premium and trustworthy tool for the real estate market.

Testing and Evaluation

The evaluation of the LiBot web and mobile-responsive platform was conducted in strict adherence to the ISO 25010 Software Quality Standards. This evaluation encompasses eight critical quality characteristics necessary for a robust software product: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. To ensure a comprehensive and highly contextual assessment, the evaluation was facilitated by a carefully selected, diverse group of respondents. This panel included IT experts specializing in software engineering and computer vision, active real estate brokers, prospective property buyers, and BSCS faculty evaluators. The selection criteria for these respondents were

predicated on their specific technical proficiencies, industry experience, and direct relevance to the system's target demographic, thereby ensuring an informed, objective, and highly practical evaluation of both the real estate marketplace features and the immersive virtual tour capabilities.

In the execution of the evaluation methodology, two distinct testing techniques were deployed to garner deep insights into the system’s overall quality and technical efficacy. The real estate brokers and property buyers utilized the Black Box Testing approach. This method focused strictly on the application’s observable inputs and outputs, specifically evaluating the ease of uploading multi-angle static room images, the seamlessness of the user interface, the visual fidelity of the resulting panoramic reconstructions, and the intuitiveness of the Marzipano-powered 360-degree viewer, all without regard for the underlying algorithmic complexities.

Conversely, the IT experts and faculty evaluators applied the White Box Testing technique, systematically delving into LiBot's internal architecture, FastAPI backend, Next.js frontend, and the core OpenCV codebase. This rigorous technical inspection allowed for a detailed examination of the system's internal logic and processing pipelines. Specifically, evaluators assessed the algorithmic accuracy of the ORB feature detection, the outlier filtration efficiency of the RANSAC homography estimation, the computational load of the image stitching process, and the application's overall adherence to robust, secure software engineering practices.

The quantitative data necessary for this evaluation was systematically gathered using a structured survey questionnaire that utilized a five-point Likert Scale. This evaluative instrument was explicitly designed to measure the system's performance, technical reliability, and overall

user satisfaction across the predefined ISO 25010 quality characteristics. The calculated responses provide a definitive and empirical understanding of LiBot's operational viability as a low-cost, high-immersion tool within the real estate sector. The statistical interpretation of these scaled results is explicitly delineated in Table

Table 1

Table of Scale and Interpretation

Scale	Interpretation
1.0 - 1.49	Strongly Disagree
1.5 - 2.49	Disagree
2.5 - 3.49	Undecided
3.5 - 4.49	Agree
4.5 - 5.00	Strongly Agree

Data Gathering Instrument

The researchers employed purposive sampling to meticulously select a targeted group of respondents, comprising IT experts, real estate brokers, property buyers, and BSCS faculty members, ensuring relevance and technical depth in the data collection process. The development of specialized questionnaires using Google Forms, tailored specifically for each distinct respondent group, facilitated the evaluation of the LiBot platform across the ISO 25010 quality standards. The evaluation focused on key criteria—namely functional suitability, usability, performance efficiency, and reliability—with multiple indicators under each criterion, such as the accuracy of the computer vision image stitching and the responsiveness of the

immersive viewer, to provide a comprehensive assessment. The validity of these instruments was rigorously confirmed by the thesis adviser and an expert panel of oral examiners during the proposal stage, underscoring the methodological severity and relevance of the study's data-gathering approach. Using the 5-point Likert scale established in Table 1, the average weighted mean is computed and interpreted as (1) 1.0 - 1.49: Very Poor, (2) 1.5 - 2.49: Poor, (3) 2.5 - 3.49: Fair, (4) 3.5 - 4.49: Good, and (5) 4.5 - 5.00: Excellent.

Data Processing and Statistical Tool

To ensure a methodical evaluation of the LiBot web and mobile-responsive platform, the researchers employed quantitative data processing techniques to analyze the responses collected through purposive sampling and Likert Scale assessments using Microsoft Excel and Google Forms. The gathered data were processed by computing the average weighted mean for each of the selected ISO 25010 Software Quality characteristics including Functional Suitability, Usability, Reliability, Performance Efficiency, Compatibility, and Security. The software tools facilitated the organization, tabulation, and coding of responses according to the scale interpretations ranging from "Strongly Disagree" to "Strongly Agree" as presented in Table II.\n\nSubsequently, the computed mean scores for each evaluation criterion provided an objective basis for determining the effectiveness, usability, reliability, and overall quality of the LiBot platform. The statistical analysis also enabled the researchers to assess the performance of the immersive virtual tour reconstruction system, particularly in terms of panorama generation, responsiveness, accessibility, and user interaction. The quantitative analysis served as a significant component in interpreting the survey results, providing a clear and systematic

evaluation of the developed system as perceived by IT experts, brokers, buyers, and faculty evaluators.